King Saud University
Muzahimya branch
College of Applied computer science

Subject: Network Protocols

ID Course: COS 370

Term: 2026 FALL

Report title: Abnormal (Lost S4 & S6)

Report No: 2

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TCP Timeline: POP3 Session, S4 and S6 Lost

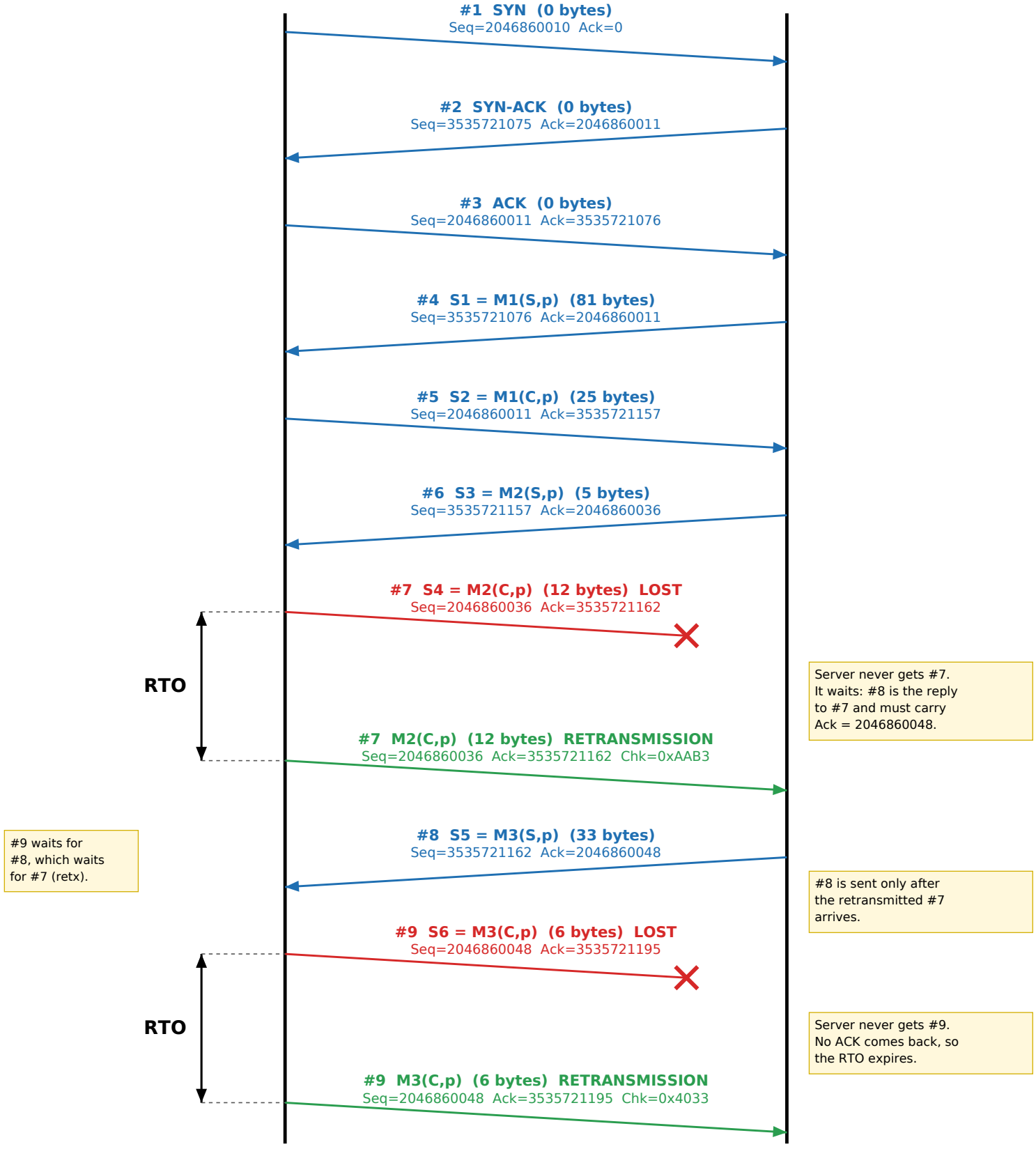
S4 = M2(C,p) #7 S6 = M3(C,p) #9

Client

P(C,p) = 1070
ISN_C = 2046860010

Server

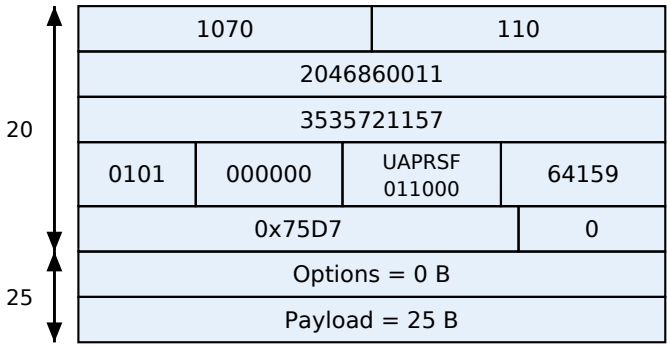
P(S,p) = 110
ISN_S = 3535721075



✕ = segment lost in transit RTO = retransmission time-out (sender waits, then resends the same segment)

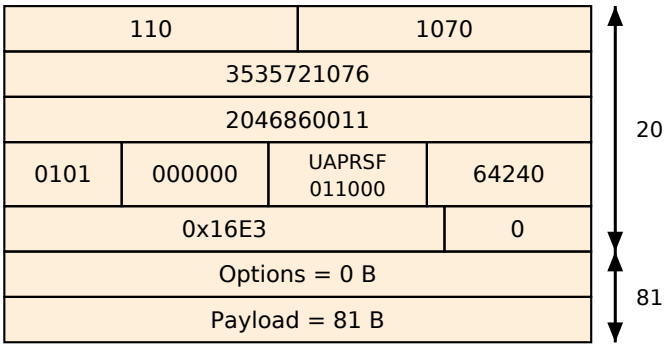
TCP Headers of the Six Data Segments

M1(C,p) #5 (25 bytes)



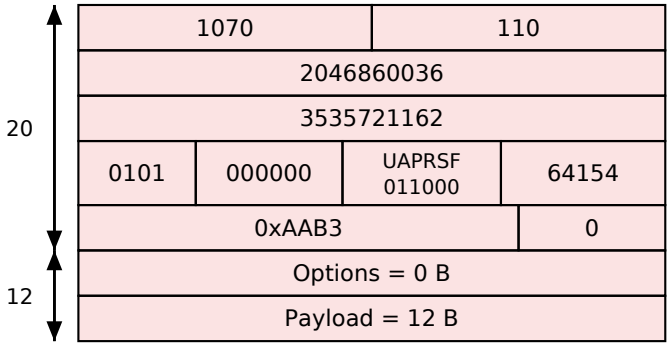
Seq = 2046860011 (= Ack of #4)
Ack = 3535721076 + 81 = 3535721157

M1(S,p) #4 (81 bytes)



Seq = ISN_S + 1 = 3535721075 + 1 = 3535721076
Ack = ISN_C + 1 = 2046860010 + 1 = 2046860011

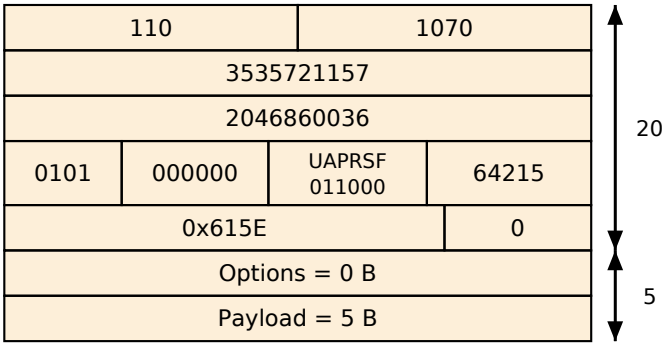
M2(C,p) #7 (12 bytes)



Seq = 2046860036 (= Ack of #6)
Ack = 3535721157 + 5 = 3535721162

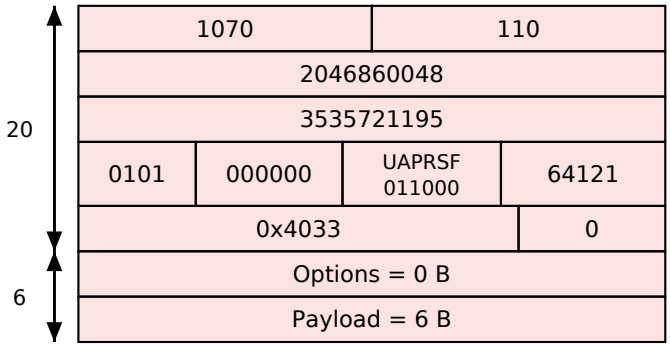
S4: LOST - the retransmission uses exactly this header.

M2(S,p) #6 (5 bytes)



Seq = 3535721157 (= Ack of #5)
Ack = 2046860011 + 25 = 2046860036

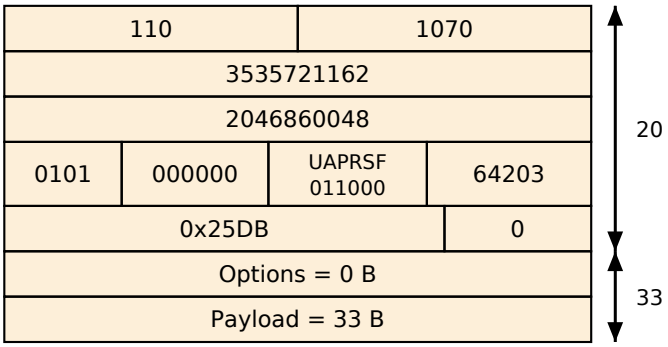
M3(C,p) #9 (6 bytes)



Seq = 2046860048 (= Ack of #8)
Ack = 3535721162 + 33 = 3535721195

S6: LOST - the retransmission uses exactly this header.

M3(S,p) #8 (33 bytes)



Seq = 3535721162 (= Ack of #7)
Ack = 2046860036 + 12 = 2046860048